Board A — Earth-side control

Dual Adafruit Perma-Proto 1/2 carrier for the LilyGo T-Display-S3 and the EC11 menu encoder. Field I/O and power leave through edge screw terminals. This board is earth-referenced only.

Draft. Header pin order along each edge is taken from the carrier with the display up and USB-C on the left. The LilyGo pinmap photo is the module bottom, so its LEFT/RIGHT labels are mirrored here. Screw terminals are numbered **T1–T24** left to right on each edge (T1 nearest USB). Outer rails are as-built. Inner rails are proposed. Column span 7–18 is assumed. On-board jumper paths are documented elsewhere.

No floating copper. Do not land `+5V_ISO`, `+12V_GD`, `−70V`, `+70V`, or `SW` on Board A. `MCU_GND` is not `−70V`.

1. Placement

View with the display facing you. USB-C is on the left. The EC11 is on the right. Pin 12 of each header is nearest USB. Screw terminals are numbered **T1–T12** on the top edge and **T13–T24** on the bottom edge, left to right. The **top** edge starts at T1 (`3V`, `GND`, one NC, then PWM, DIR, limits, I^{2}C). The **bottom** edge starts at T13 (`5V`, `GND`, five NC including T17–T19, then OPEN, LIM-L, ISENSE, HALL, 3V). The EC11 uses GPIO11–13 on the T-Display header directly; see EC11 on-board wiring.

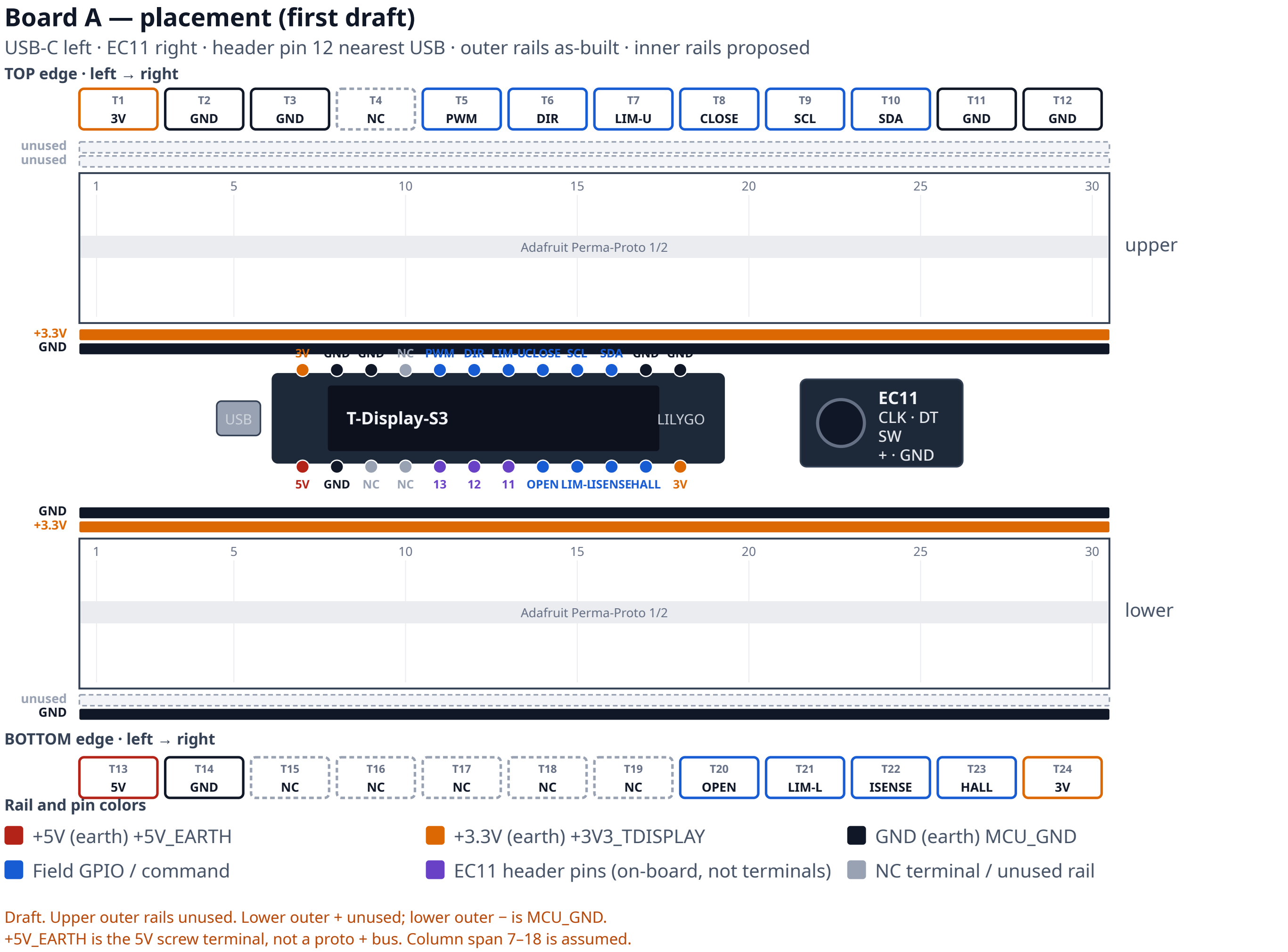


Figure 1. Grid-true placement. Terminal IDs T1–T24 label each screw block left to right. Dashed gray bars are unused rails (as-built: both upper outer buses, and the lower outer +). Purple dots are T-Display header pins wired to the on-board EC11 (GPIO11–13). T17–T19 are NC; the encoder does not use those terminals.

2. Rails

Each half has four full-width buses. On the as-built unit both **upper outer** rails are unused, and the **lower outer +** rail is unused. The **lower outer −** rail is `MCU_GND`. Inner-rail nets below are still proposed. `+5V_EARTH` lands on the **5V** screw terminal and the T-Display **5V** pin; it is not on a proto + bus. Sensors / EC11 + should see `+3V3_TDISPLAY` from the module **3V** pin. Do not back-feed 3.3 V into the module.

Rail	Net	Diagram color
Upper outer −	unused (as-built; no jumper)	■ unused
Upper outer +	unused (as-built; no jumper)	■ unused
Upper inner +	`+3V3_TDISPLAY` (proposed)	■ +3.3V (earth)
Upper inner −	`MCU_GND` (proposed)	■ GND (earth)
Lower inner −	`MCU_GND` (proposed)	■ GND (earth)
Lower inner +	`+3V3_TDISPLAY` (proposed)	■ +3.3V (earth)
Lower outer +	unused (as-built; no jumper)	■ unused
Lower outer −	`MCU_GND` (as-built)	■ GND (earth)

3. What attaches

Power

Board A runs from the earth-side Mean Well **HDR-15-5 #1** supply (`+5V_EARTH` / `MCU_GND`). Land the PSU positive on **T13** (`5V`) and the PSU negative on **T14** (`GND`). Those terminals feed the T-Display **5V** pin and the board `MCU_GND` reference. USB-C on the module remains for bench programming. Do not back-feed 3.3 V into the module **3V** pin.

Screw terminals

Numbering follows Figure 1: left to right with the display facing you and USB-C on the left. T1 and T13 are nearest USB.

Top edge — T1–T12

Terminal	Label	GPIO / net	Attaches
T1	**3V**	`+3V3_TDISPLAY`	SS49E VCC, AS5600 VCC (from module **3V**)
T2	**GND**	`MCU_GND`	Sensor, limit, and rocker returns
T3	**GND**	`MCU_GND`	Sensor, limit, and rocker returns. Lands **R6** (PWM) and **R7** (DIR) 10 kΩ pull-downs
T4	**NC**	—	Module NC. Unassigned. May be jumpered to `MCU_GND`.
T5	**PWM**	GPIO16	Board B 6N137 LED. On-board **R6** 10 kΩ to T3 `MCU_GND`
T6	**DIR**	GPIO21	Board B U4 LED. On-board **R7** 10 kΩ to T3 `MCU_GND`
T7	**LIM-U**	GPIO17	Upper NC limit to GND
T8	**CLOSE**	GPIO18	Close rocker (active-low, paralleled pair)
T9	**SCL**	GPIO44	AS5600 SCL
T10	**SDA**	GPIO43	AS5600 SDA
T11	**GND**	`MCU_GND`	Sensor, limit, and rocker returns; Board B LED return
T12	**GND**	`MCU_GND`	Sensor, limit, and rocker returns

Bottom edge — T13–T24

Terminal	Label	GPIO / net	Attaches
T13	**5V**	`+5V_EARTH`	**Power in.** Mean Well HDR-15-5 #1 (+) → T-Display **5V**
T14	**GND**	`MCU_GND`	**Power in.** Mean Well HDR-15-5 #1 (−) / earth-side return
T15	**NC**	—	Module NC. Unassigned.
T16	**NC**	—	Module NC. Unassigned.
T17	**NC**	—	Unassigned. Same index as GPIO13; EC11 SW is on-board from the header (EC11 wiring).
T18	**NC**	—	Unassigned. Same index as GPIO12; EC11 DT is on-board from the header.
T19	**NC**	—	Unassigned. Same index as GPIO11; EC11 CLK is on-board from the header.
T20	**OPEN**	GPIO10	Open rocker (active-low, paralleled pair)
T21	**LIM-L**	GPIO3	Lower NC limit to GND
T22	**ISENSE**	GPIO2	ACS712-05B OUT after 10 kΩ/20 kΩ (≤ 3.3 V)
T23	**HALL**	GPIO1	SS49E OUT
T24	**3V**	`+3V3_TDISPLAY`	SS49E VCC, AS5600 VCC (from module **3V**)

EC11 (on-board)

The menu encoder wires directly to T-Display-S3 header pins. It does not use screw terminals. Terminals **T17–T19** are NC on the as-built unit.

EC11 breakout	T-Display pin	GPIO / net
CLK	LEFT GPIO11	GPIO11
DT	LEFT GPIO12	GPIO12
SW	LEFT GPIO13	GPIO13
+	**3V**	`+3V3_TDISPLAY`
GND	**GND**	`MCU_GND`

Full notes, terminal clarification, and links: EC11 on-board wiring.

On-board pull-downs

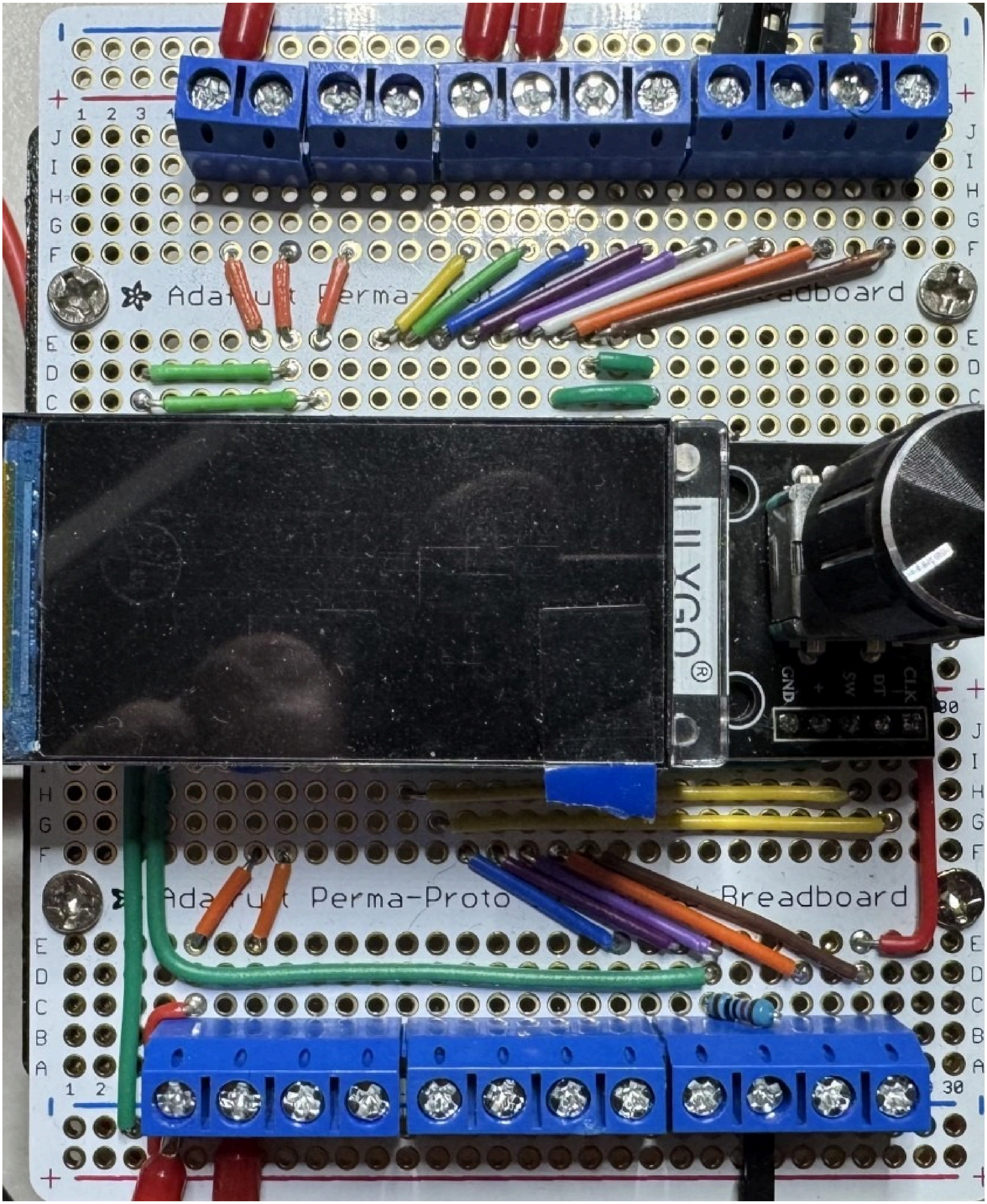
R6 (10 kΩ) sits between **T5** (`PWM`, GPIO16) and **T3** (`MCU_GND`). **R7** (10 kΩ) sits between **T6** (`DIR`, GPIO21) and the same T3 return. They hold the Board B optocoupler LEDs off if the T-Display is unpowered or resetting. Series LED resistors stay on Board B.

Board A → Board B

Keyed 3-pin earth-side cable from **T5** (`PWM`), **T6** (`DIR`), and a `MCU_GND` terminal (for example **T11** or **T14**). LED resistors stay on Board B.

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Wired carrier. USB-C on the left, EC11 on the right.

Related

This page is a chapter of the Wiring Guide. Electrical wiring truth for the motor stack remains the gate-driver schematic, not the placement drawing.